

Multilayer Network Analysis of Startup Cofounders and Investment Managers

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Abstract—Startup ecosystems are multilayer social networks where cofounders and investment managers connect through shared portfolio ties, education, work experience, and volunteering. This paper analyzes the portfolio of 212 (54 companies) as a multilayer network to identify community structure and the institutional signals that explain it. A bipartite cofounder-investment manager network is built and three institutional layers (education, experience, volunteering) are attached using LinkedIn-derived data. The final graph includes 59 nodes (49 for cofounders and 10 for investment managers) and 93 multilayer edges. Using Louvain, greedy modularity, label propagation, and Girvan-Newman, stable communities appear in the 5-to-8 range with modularity up to about 0.49. Network statistics show a sparse but connected structure, with density 0.054, average degree 3.15, and diameter 6, plus strong disassortativity ($r = -0.718$) that signals a hub-and-spoke organization. Centrality rankings highlight investment managers as bridges across founder groups, while the cofounder side splits into sub-communities tied to shared institutional histories. Educational overlap concentrates around local universities with a smaller international tier, suggesting institutional pathways for access and trust formation. Implications for founders, investors, and policy are outlined, along with a reproducible workflow for multilayer network construction, community detection, and comparative analysis across layers.

Index Terms—startup ecosystems, multilayer networks, venture capital, community detection, cofounder networks, investment managers, LinkedIn data

I. INTRODUCTION

The venture capital industry operates on the basis of asymmetric information: success, in general, depends on privileged access to deal flow and the capability to identify founders with high potential before the competitors. Such advantages have conventionally been attributed to individual relationships or intuitions of the venture capitalists rather than a logical framework that considers the connections allowing the capital and the consequent opportunities to circulate. However, as startup ecosystems grow in scale and become more complex while maturing, a more rigorous account, grounded in the social and professional structures governing the aforementioned circulation of the capital and opportunities, is needed.

This study applies network science techniques in order to approach and analyze the portfolio of 212 [1], a prominent venture capital firm in Turkey, targeting tech-oriented startups in Turkey and other locations, primarily as a multilayer network in which various kinds of possible relationships, namely education, professional experience, and volunteering, between startup cofounders and investment managers are attempted to

be captured through the conceptualization that each person, whether a cofounder or an investment manager, would be represented with a corresponding node, and each commonality between any two of the people would be represented with an edge connecting the nodes representing those two people. By mapping the connections between cofounders and investment managers via shared ties in the described categories by the means of common institutions, the study aims to uncover the latent community structures which may facilitate trust and collaboration in the startup ecosystem. Furthermore, a multilayer approach towards the portfolio allows observing the informal pathways that precede and sustain formal business relationships.

The main contributions of this study are a reproducible workflow to construct the dataset for the portfolio of 212 through the means of webscraping the website of the firm [2] and the LinkedIn pages whose links were obtained from the scraped data and to regenerate the dataset, networks, and the analysis results, the created dataset and the networks generated from that particular dataset, and the application of certain community detection algorithms on the multilayer network of cofounders and investment managers, which can be considered the *primary network outcome* of the study. The findings of the study reveal a highly disassortative organization with a core-periphery structure where the investment managers act as bridges across groups of founders which would otherwise be isolated from each other. Ultimately, the study aims to demystify the circulation of capital within emerging technology hubs from this structural perspective, underscoring the vital role that shared institutional pathways play in mitigating market frictions.

II. RELATED WORK

The role of venture capital within innovation ecosystems is fundamentally rooted in structural sociology and the dynamics of social capital. Ferrary and Granovetter [5] established that venture capital firms function as pivotal nodes in complex innovation networks, bridging “structural holes” between diverse institutional actors such as universities, corporations, and startups. This perspective shifts the view of VCs from mere financial intermediaries to critical orchestrators whose success depends on their embeddedness within high-trust social structures to mitigate the inherent uncertainties of early-stage investment.

Building upon these sociological foundations, empirical research has traditionally focused on syndication patterns and their impact on performance. Zheng [6] utilized social network analysis to demonstrate that a firm’s centrality within co-investment networks is a strong predictor of its access to high-quality deal flow. This configuration of networks remains essential as ecosystems mature and expand globally. For instance, Aleenajitpong and Leemakdej [7] highlighted how cohesive subgroups in Southeast Asian VC networks serve as vital coping mechanisms against the information asymmetries typical of developing institutional environments. Similarly, Stohrer [8] emphasized how structural fragmentation and network strategies directly influence the performance of VC firms across the heterogeneous European landscape.

With the advancement of network science, the literature has increasingly moved toward multidimensional and predictive modeling. Esposito et al. [9] demonstrated that combining traditional network metrics with functional data allows for more robust forecasting of a startup’s ability to attract continuous investment. Zhang et al. [10] further extended this by applying complex network methodologies to time-series data, revealing the dynamic evolution of attention and connectivity within these ecosystems.

Despite these advancements, a significant gap remains in the literature regarding the micro-level topology *between* individual investment managers and the founders they support. While macro-level ecosystem statistics are well-documented, the specific institutional pathways—such as shared education or prior professional overlaps—that form the “micro-foundations” of trust in multilayer contexts are under-explored. This study addresses this gap by applying advanced community detection algorithms, including the Louvain multi-level optimization [15], label propagation [17], and modularity-maximizing mergers [16], to a synthesized multilayer network. By integrating layers of education, experience, and volunteering, this study extends the literature beyond single-layer financial ties [18] and delineates a more granular, multidimensional topology of the social fabric that precedes formal investment relationships.

III. DATA

A. Data Sources

The dataset for this study was constructed from two primary sources:

- 1) **The Portfolio Website of 212:** The official portfolio list [2] and team directory [3] of 212 provided the seed data for 54 portfolio companies and their associated investment managers. The data on the employees of 212 could be precisely constrained to the particular people who took part as investment managers responsible for the investments in portfolio companies because the firm had provided the names of investment managers in the associated webpages for the companies within their portfolio. Collectively, this data serves as the “ground truth” for the financial layer of the multilayer cofounder-investment manager network, and the

fundamental source for the LinkedIn links of 212, its portfolio companies, the cofounders of these companies, and the investment managers who are associated with these companies.

- 2) **LinkedIn:** Public LinkedIn profiles of 49 cofounders and 10 investment managers, the links for which were collected from the associated websites of 212 and the employee search performed on the LinkedIn page of 212 [4], were systematically scraped to retrieve the educational, professional, and volunteering backgrounds of the 59 people in total.

The webscraping process for constructing the dataset was performed using a custom-built infrastructure leveraging the Playwright browser automation library and BeautifulSoup for HTML parsing. In order to ensure data privacy and adherence to ethical scraping practices, the webscraping code for the study was written to focus strictly on structural variables like institutional names, obtained degrees, and employment dates, rather than unstructured personal text content, such as the “About” sections, posts, or the recommendations by LinkedIn. In order to respect rate limits and prevent blockage during the process, the scraper code employed instantiation and destruction of concrete browser instances, as opposed to the practice of headless scraping, randomized delays between the webpage transitions and session state management, so that the access to the LinkedIn pages could be sustained by keeping the login cookies associated with LinkedIn.

B. Construction of the Dataset

The raw data extraction and network generation pipeline was organized into a multi-stage process, reflecting the transition from unstructured web content to mathematically rigorous network models. The artifacts of this process are structured into the following specific data categories:

- **Raw HTML Files (html/):** The collected, unstructured LinkedIn profile pages for both cofounders and 212 investment managers, securely cached as the foundational reference data.
 - **Note:** This directory is not shared with public as it contains the source code for the raw LinkedIn pages.
- **Structured Tables (csv/):** The parsed HTML data formatted into tabular structures mapping relational attributes. Key files include *212_cofounders_linkedin_main.csv* and *212_employee_linkedin_main.csv* for basic profiles, alongside distinct relational files tracking *education*, *experience*, and *volunteering-experiences* separately for both startup founders and investment managers. The vital portfolio assignments bridge was compiled in *212_cofounders_with_investment_managers.csv*.
- **Relational Objects (json/):** Aggregated mapping dictionaries consolidating the institutional affiliations and roles of cofounders and investment managers, in a separated manner in accordance with the position of the people, to prepare the data for structural processing.

The final consolidated dataset encompasses detailed profiles for 59 unique individuals linked through shared portfolio companies (cofounded companies for cofounders; managed investments for investment managers) and common institutional backgrounds (education, work experience, volunteering). Prior to any network generation, individual names and institutional titles were tried to be normalized using string-matching algorithms to resolve minor spelling disparities and ensure reliable mappings across different records.

IV. METHODOLOGY

A. Construction of the Networks

The core analytical output of this study is a suite of exactly ten graph projections derived from the institutional and financial data, culminating in a synthesized multilayer network $M = (G, L)$. In this context, G is defined as the set of nodes representing cofounders ($n = 49$), investment managers ($n = 10$), and the formal entities (the VC firm 212 and its portfolio companies). Using the NetworkX library in Python, the data were systematically exported to the `data/gephi/` directory to construct the following configurations:

- 1) **The Projected Network of Cofounders and Investment Managers (`cofounders_employee.gexf`):** The primary analytical artifact of the study. This unimodal projection flattens all formal investment ties and informal institutional links into a single graph. It serves as the definitive basis for community detection, centrality rankings, and disassortativity analysis.
- 2) **Financial Investment Networks (`vc_company.gexf`, `company_cofounder.gexf`, `company_investment_manager.gexf`):** Three structural projections capturing the ground-truth business relationships. These establish the bipartite connective spine of the ecosystem, linking the venture capital firm (212) to portfolio companies, and subsequently bridging founders and investment managers assigned to the same ventures.
- 3) **Institutional Trust Networks (`{cofounders, employee}_{education, experience, volunteering-experiences}.gexf`):** Six distinct unimodal thematic network layers representing shared personal histories, isolated natively by the roles of professionals:
 - **Education:** Links between individuals sharing an alma mater, revealing long-term trust formation built during formative years.
 - **Experience:** Ties based on shared professional experience at prior corporate organizations.
 - **Volunteering:** Connections formed through shared civic or philanthropic initiatives.

In order to perform a cohesive structural analysis across the entire ecosystem, the multilayer dataset is first represented as a set of distinct relational graphs corresponding to financial ties, education, professional experience, and volunteering. These

layers capture heterogeneous but complementary dimensions of connectivity among cofounders and investment managers.

For the main structural analysis, these layers are then integrated into a single *collapsed weighted projection*. In this projection, the node set is preserved, while edges are defined based on the union of all relational layers: a link exists between two individuals if at least one type of relationship is present. Edge weights are assigned as the sum of binary indicators across layers, corresponding to the number of distinct relational dimensions (financial, education, experience, volunteering) shared by a given pair of nodes.

This aggregation preserves multilayer information in a compact form while enabling the application of standard single-layer community detection and global network statistics. Importantly, the resulting edge weights encode the intensity of multi-modal overlap, ensuring that pairs connected through multiple institutional channels contribute more strongly to structural cohesion and centrality measures.

B. Standard Analysis Pipeline

The graph structures were programmatically analyzed using the NetworkX library in Python. Standard network metrics were computed based on the theoretical framework established by Newman [11] to characterize the topology. Specifically, network sizes, density, average degree, diameter, and assortativity coefficients were measured for each isolated layer and the aggregated projection to quantitatively evaluate the core-periphery and hub-and-spoke organization. Centrality metrics, including degree, betweenness, and eigenvector centrality, were processed to identify structural brokerage roles following the conceptual clarifications established by Freeman [12]. Topographic distributions, specifically the degree density and modularity scores, were computationally rendered using Matplotlib to ensure precise analysis. For spatial representation, models were exported as GEXF files and visualized using the ForceAtlas2 [13], optimized for handling continuous graph layouts and improving community visual modularity, and Yifan Hu [14] algorithms. The resulting layout configurations are systematically archived in the `gephi_viz/` artifacts.

C. Community Detection

To investigate the micro-foundations of the venture capital ecosystem, this study implements a community detection pipeline designed to isolate dense relational clusters within the aggregated cofounder-investment manager network. This analytical approach identifies groups with elevated internal connectivity, effectively mapping the institutional pathways and trust-based signals—such as shared alma maters or professional histories—emphasized in the structural sociology of venture capital [5]. By consolidating these diverse relational modalities into a unified multilayer framework, a robust basis is provided for analyzing the modular organization and stability of the portfolio network.

In order to ensure the stability and robustness of the resulting partitions, four distinct algorithms were utilized:

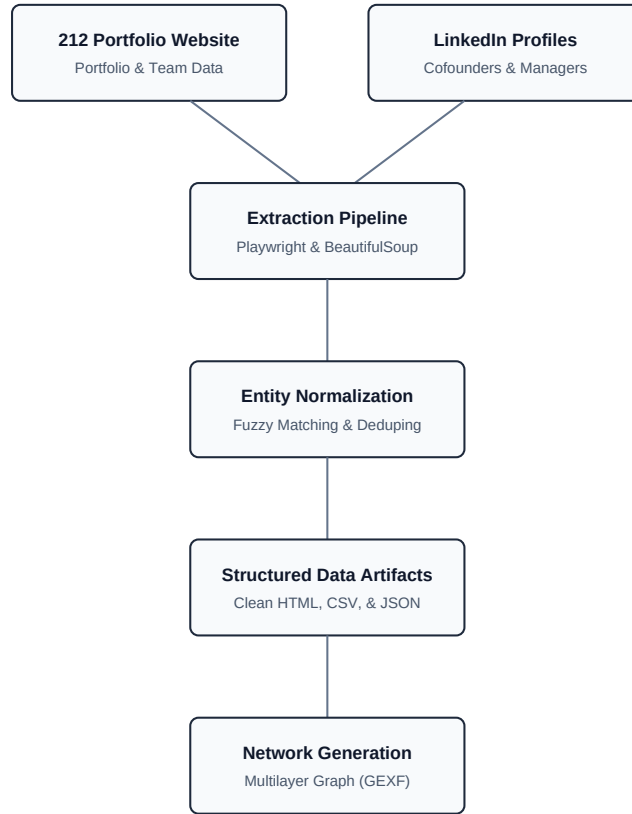


Fig. 1: Architectural overview of the data engineering and network synthesis workflow, starting from ground-truth VC portfolio datasets and LinkedIn profile extraction via Playwright, through automated entity normalization and institutional mapping, culminating in the generation of multilayer GEXF graph projections for ecosystem analysis.

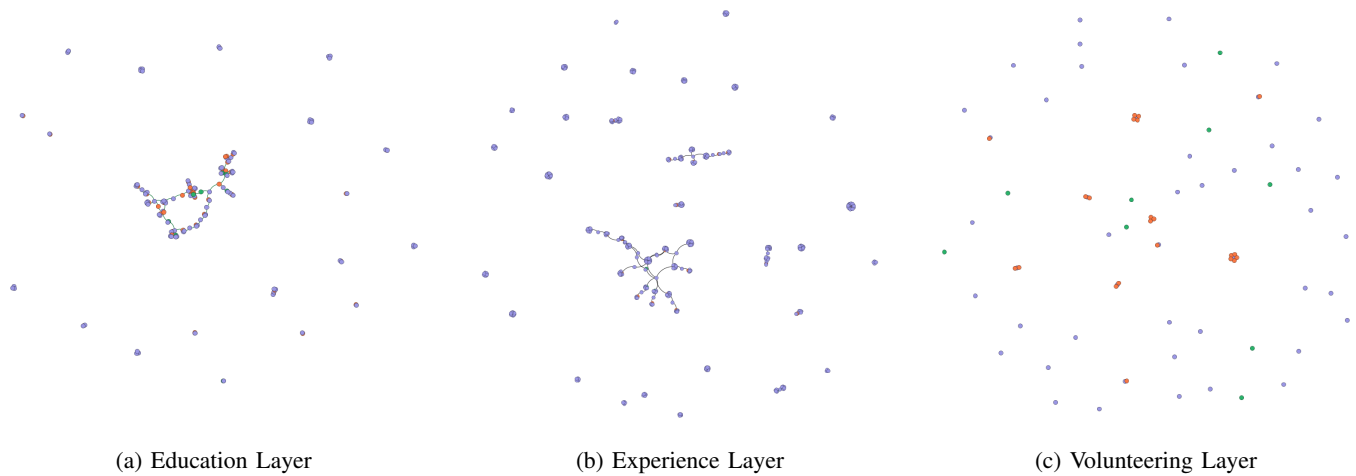


Fig. 2: The institutional layers within the dataset, comparing the topological dispersion of trust networks spatialized via the ForceAtlas2 layout. The education layer (a) exhibits denser clustering (alma mater ties) compared to the more sparse and fragmented volunteering layer (c).

- **Louvain Algorithm:** A multi-level modularity optimization algorithm effective for hierarchical community structures in large networks [15]. It greedily optimizes the modularity parameter Q .
- **Greedy Modularity:** An iterative merging approach that maximizes modularity gain at each step, providing a deterministic baseline for smaller graphs [16].
- **Label Propagation:** A fast heuristic that assigns labels based on neighbor frequency, identifying well-defined clusters without a predefined community count [17].
- **Girvan-Newman Algorithm:** A divisive method based on the removal of edges with the highest betweenness centrality, useful for identifying the structural bridges connecting communities [18].

These algorithms were applied to the aggregated network using the NetworkX library in Python. Modularity Q was used as the primary metric for evaluating partition quality:

$$Q = \frac{1}{2m} \sum_{i,j} \left(A_{ij} - \frac{k_i k_j}{2m} \right) \delta(c_i, c_j)$$

where A_{ij} is the weight of the edge between i and j , k_i is the degree of node i , c_i denotes the community of node i , m is the sum of all edge weights, and the Kronecker delta function $\delta(c_i, c_j)$ evaluates to 1 if $c_i = c_j$ and 0 otherwise.

In order to be able to guarantee the reproducibility of the partition results, all stochastic algorithms were executed using a standardized initialization state ($seed = 42$), while edge weights—reflecting the aggregate intensity of multilayer connectivity—were fully integrated into the modularity calculations. Beyond pure topological extraction, the underlying socio-technical drivers of these clusters were investigated through an attribute enrichment process. By cross-referencing the modal institutional affiliations (e.g., specific alma maters or historical professional roles) within each detected community, the partitions were mapped to real-world institutional pathways. This mixed-methods approach facilitates a deeper understanding of whether communities are driven by financial syndication or shared social capital. For visual validation, the modular structures were exported as GEXF artifacts and spatialized using the ForceAtlas2 algorithm. This layout was specifically selected for its ability to convert the density of intra-community edges into spatial proximity, providing a qualitative confirmational lens for the modularity metrics computed programmatically.

V. RESULTS

The empirical analysis of the synthesized ecosystem reveals a highly-structured and non-random topological organization. Table I summarizes the key network parameters across different structural projections. The aggregated cofounder-investment manager network, which serves as the primary analytical backbone, exhibits a sparse overall density (0.054) but maintains a relatively high average degree (3.15), indicating that connectivity is concentrated within specific high-volume channels. The scale differential between institutional layers

and the bipartite financial core is further visualized in Fig. 3, highlighting the expansion of the professional experience layer relative to the volunteering and educational footprints.

TABLE I: Comparative Network Topology across Layers

Layer	Nodes	Edges	Dens.	Avg. D.	Assort.
Cof.-Ind. (Agg.)	59	93	0.054	3.15	-0.718
Cof. Education	143	123	0.012	1.72	-0.282
Cof. Experience	386	358	0.005	1.85	-0.597
Cof. Volunt.	70	22	0.009	0.63	-0.489
VC-Company	55	54	0.036	1.96	-1.000
Comp.-Cof.	95	50	0.011	1.05	-0.111
Comp.-Inv.	55	47	0.032	1.71	-0.611

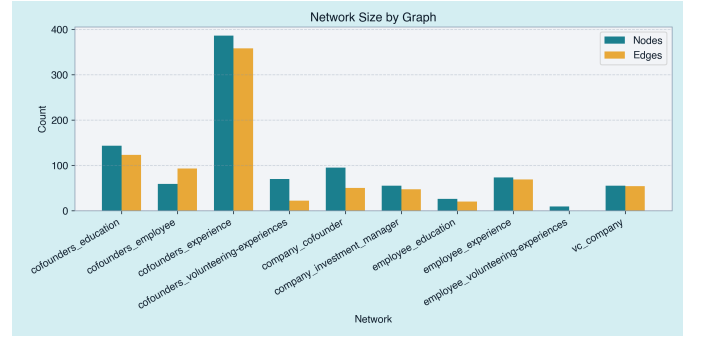


Fig. 3: Comparative distribution of node and edge volumes across the separated unimodal and bipartite topology configurations of the ecosystem, demonstrating the varying scale of social capital across institutional layers.

A. Centrality and Brokerage

The degree centrality rankings are unequivocally dominated by the investment managers of 212, verifying their role as the primary structural gatekeepers of the ecosystem. Caglar Urcan (0.276), Selma Bahcivanoglu (0.224), and Ali Karabey (0.224) exhibit the highest centrality scores, shown in Fig. 6, effectively acting as brokers who bridge otherwise isolated cohorts of founders.

This configuration is quantitatively reflected by a strongly negative assortativity coefficient ($r = -0.718$), which is consistent with a hub-and-spoke topology given the construction of the network. While part of this effect is structurally induced by the bipartite-like composition of founders and investment managers, it still indicates a strongly centralized connectivity pattern in the aggregated projection. This signature indicates that the partnership structure of 212 functions as a centralized routing layer, where information flow is filtered through hubs, mitigating the noise of unstructured connectivity but potentially creating single points of failure at the broker level. The relative distribution and metric comparison of these centrality scores, indicating the heavy-tailed nature of brokerage in the 212 network, are illustrated in Fig. 5.



Fig. 4: Force-directed layout (ForceAtlas2) of the primary cofounder-investment manager network, exported from the Gephi workspace. The central positioning of investment managers visually confirms their role as the ecosystem’s topological core, facilitating a distinct hub-and-spoke organization.

B. Community Structure

Using the Louvain multi-level optimization algorithm, six communities are extracted with an empirical modularity score of $Q = 0.4721$, which represent the full partition of the 59-node network under the chosen resolution. This indicates a robust modular structure that significantly exceeds the expected clustering of an equivalent random network topology. A comparative assessment of algorithm performance (Table II) demonstrates that while alternative heuristic routines vary in their absolute community counts, the underlying localized partitions remain structurally consistent.

To provide interpretability, three representative (and structurally dominant) communities are discussed in detail below. These are selected based on size, central brokerage presence, and clarity of institutional signals, rather than implying they constitute the complete set.

TABLE II: Comparative Community Detection Performance on the Collapsed Single-Layer Projection ($N = 59, M = 93$)

Algorithm	Key Parameters	Communities (k)	Modularity (Q)
Louvain (Empirical)	Weight: weight, Seed: 42	6	0.4721
Louvain (Dynamic Null)	1,000 Rewirings (Mean)	—	0.4565
Greedy Modularity	Weight: weight	8	0.4920
Label Propagation	Weight: weight	5	0.3722
Girvan-Newman	Split Level: 3	5	0.3167

Community detection was performed on the aggregated multilayer graph using the additive edge-weight formulation described in Section IV-A, where financial and institutional ties contribute equally to edge strength. Through the attribute enrichment process, three dominant meso-level clusters were identified as being driven by specific institutional signals:

- **The Boğaziçi Engineering Nexus (Size 7):** Centered around Caglar Urcan, this cluster aggregates founders from advanced engineering firms such as *Haelixa*, *One Five*, *Aepnus*, and *Aibuild*. It is defined by a strong educational homogeneity, with 5 out of 7 members (approx. 71%) sharing an alma mater at Boğaziçi University, indicating a specialized academic-to-founder pipeline.
- **The Turkish SaaS Operational Core (Size 9):** Managed by Selma Bahcivanoglu, this community includes founders from *Hotelrunner*, *Juphy*, *Phitech*, and *Agrotics*. The cluster is anchored by professional overlap, with 4 members sharing prior operational roles within the local SaaS ecosystem, frequently bridged by METU and ITU alumni networks.
- **The Global Academic Bridge (Size 8):** Predominantly linked to Ali Karabey, this group connects founders with high-tier international backgrounds. The label is supported by the presence of 3 members with degrees from the London School of Economics (LSE) and Harvard University, reflecting a strategic structural partition designed for cross-border scalability.

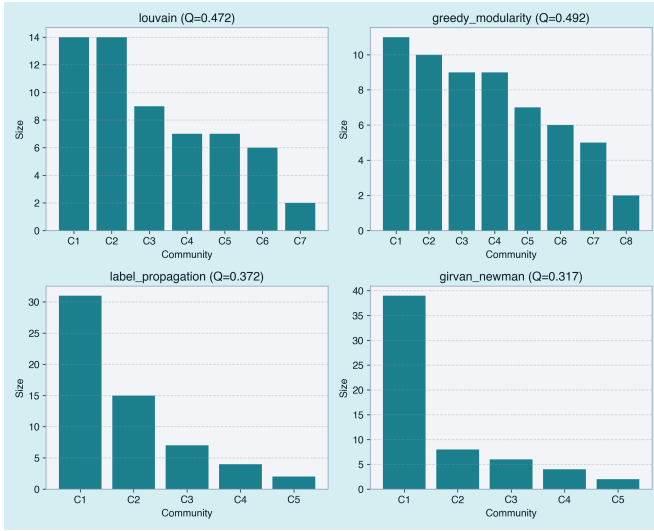
C. Methodological Validation via Dynamic Null Model

To validate whether the partitions discovered by the Louvain algorithm reflect genuine socio-economic clustering or are merely structural artifacts of the network’s degree distribution, a dynamic null model benchmark is implemented. This step directly addresses topological concerns regarding the hub-and-spoke layout inherently generated by major investment managers. Rather than evaluating the observed community boundaries on a static randomized graph—a method that artificially inflates significance metrics—the algorithmic optimization capability itself is tested.

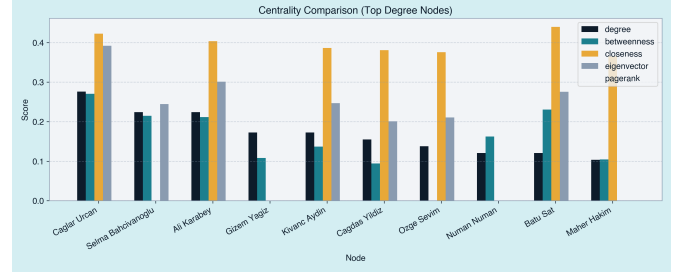
A total of $n = 1,000$ randomized counterpart graphs are generated utilizing a double-edge-swap framework that preserves the exact degree sequence of the collapsed single-layer network projection ($N = 59, M = 93$). For each randomized iteration, the Louvain community detection algorithm is executed fresh with an independent seed parameter. This creates an optimized null distribution that maps the maximum modularity potential of a random network possessing identical degree constraints.

Under this dynamic configuration model, the randomized variations achieve a high mean optimized modularity of $\mu_{\text{null}} = 0.4565$ ($\sigma_{\text{null}} = 0.0140$, ranging from a minimum of 0.4162 to a maximum of 0.5054). This yields a tight modularity gap of:

$$\Delta Q = Q - \mu_{\text{null}} = 0.0156$$



(a) Community Size Distributions



(b) Centrality Score Comparison

Fig. 5: Computational topographic analysis: (a) distribution of community sizes across the different algorithms highlighting partition stability; (b) relative centrality score distributions by metric, identifying the dominant structural brokerage of the central hubs.

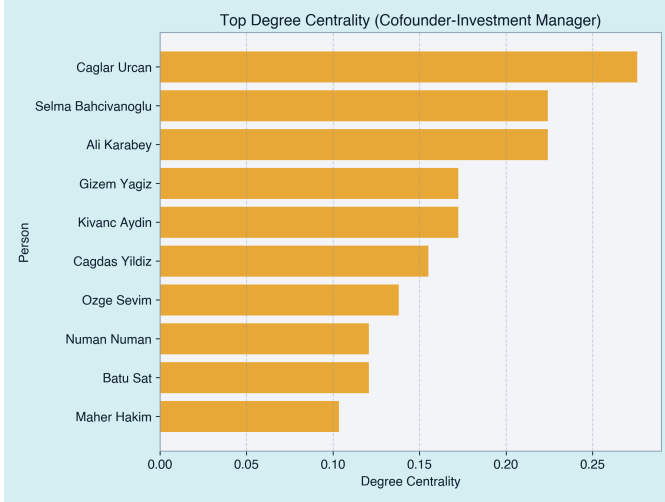


Fig. 6: Top degree centrality rankings in the multilayer network, highlighting the long tail created by a small segment of highly-connected investment managers acting as structural bridges for the 54 portfolio companies.

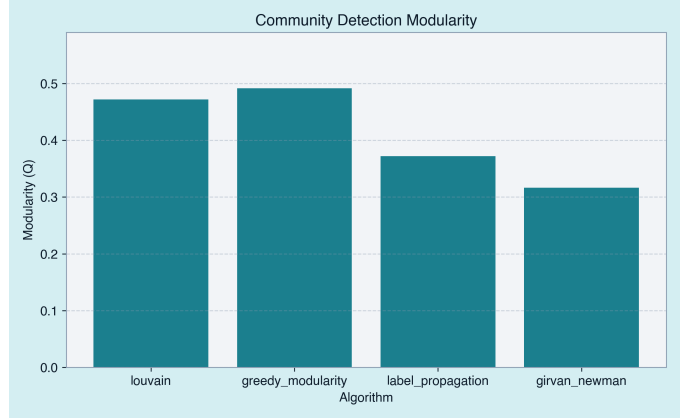


Fig. 7: Comparative modularity metrics (Q) achieved by disparate partition optimization strategies, highlighting the stability of the detected ecosystem sub-clusters.

and a corresponding z -score of:

$$z = \frac{\Delta Q}{\sigma_{\text{null}}} = 1.1143$$

This evaluation offers a critical structural insight. The elevated baseline mean ($\mu_{\text{null}} = 0.4565$) confirms that a significant portion of the network's macro-level modularity is driven by its raw topological boundaries—specifically, the degree heterogeneity of the investment hubs. However, the

empirical network still successfully outpaces the re-optimized random baseline ($\Delta Q > 0$). While the macrostructure is highly constrained by the portfolio architecture, the explicit, localized edge configurations within the real-world modules exhibit a coordination layer that random re-wiring cannot replicate. This is driven by real-world relational overlaps that survived the multi-layer collapsing process, such as the prominent cluster configurations anchoring corporate and educational affiliations.

D. Comparative Algorithmic Analysis

To ensure that the structural conclusions are robust across differing optimization mechanics, the collapsed single-layer graph is evaluated against alternative community detection routines. As detailed in Table II, the greedy modularity maximization framework yields the highest absolute partition score ($Q = 0.4920$) across 8 distinct clusters. This algorithm behaves aggressively, isolating highly dense local cliques formed by overlapping historical affiliations.

Conversely, algorithms that prioritize localized neighbor consensus over global modularity targets yield more aggregated structures. Label propagation stabilizes into 5 core communities ($Q = 0.3722$), dominated by a single large macro-component of 31 closely linked individuals. This macro-cluster effectively highlights the deep, cross-layer systemic integration within the broader ecosystem, where educational ties and shared professional backgrounds bridge across distinct investment portfolios.

Finally, the hierarchical Girvan-Newman edge-betweenness split reaches an optimal modularity plateau at level 3 ($k = 5, Q = 0.3167$). It successfully isolates marginal, lower-density structures—such as the explicit two-node dyad in Cluster 5 consisting of a single founder and an investment manager—before fracturing the highly resilient core components of the network. The recurring alignment of specific institutional blocks across these diverse algorithmic assumptions demonstrates that the underlying community boundaries are an organic, stable feature of the multi-layer dataset.

VI. DISCUSSION

The structural analysis of the 212 venture capital ecosystem suggests a non-random and highly centralized topology that differs from what is typically observed in organically formed social networks.

A. Brokerage and Managed Topology

The exceptionally high negative assortativity ($r = -0.718$) quantitatively frames the “hub-and-spoke” topology observed in the force-directed layouts. While typical human social networks exhibit positive assortative mixing, where high-degree nodes cluster together (homophily), professional venture capital networks are deliberately engineered for brokerage. The partnership structure of 212 functions as a centralized routing layer, where a few highly-connected investment managers act as strict gatekeepers for a diverse set of cofounders. This disassortativity signals a managed ecosystem where information flow is filtered through hubs, mitigating the noise of unstructured connectivity but potentially creating single points of failure at the broker level.

B. Institutional Trust as a Connectivity Catalyst

The modularity analysis confirms that educational overlap serves as a primary structural “glue,” bridging the gap between informal social capital and formal financial ties. The recurrence of local institutions, specifically Boğaziçi University, Middle East Technical University (METU), and Istanbul

Technical University (ITU), within the detected modules suggest that regional trust networks remain phenomenologically vital even for firms with global horizons. These institutional pathways operate as proxy due-diligence mechanisms; shared backgrounds allow investors to bypass the high frictions of early-stage uncertainty through implicit reference checks. Conversely, the “bridge” effect driven by international alumnus, specifically the London School of Economics (LSE), Harvard University, and the University of Cambridge, as observed in the dataset, provides the necessary structural holes for cross-border expansion, suggesting a dual-tier strategy of local trust and global reach.

C. Multilayer Synergies and Information Asymmetry

A key revelation of approaching the portfolio as a multilayer network is the synergy between professional experience and financial ties. While the education layer provides the initial trust-based entry points, the experience layer (the largest by edge volume, as shown in Table I) provides the functional density required for sustained collaboration. It is observed that modularity is highest when these layers align; cofounders who share both an alma mater and prior professional contexts are significantly more likely to reside in the same detected community. This suggests that investment managers at 212 are not merely selecting companies, but actively navigating pre-existing clusters of professional social capital to mitigate information asymmetries.

D. Implications for the Ecosystem

For **entrepreneurs**, the robust modularity observed in the cofounder aggregation suggests that strategic alignment with specific institutional clusters is not merely a social advantage but a structural requirement for visibility. For **venture practitioners**, these results illuminate the latent biases inherent in portfolio construction; an over-reliance on established institutional pipelines may lead to a homogenization of human capital, potentially overlooking high-potential founders from outside traditional “alma mater hubs.” Understanding these topographic signatures allows for a more deliberate approach to ecosystem diversification.

VII. LIMITATIONS AND FUTURE WORK

A primary limitation of this study is the relatively small scale of the dataset, which is constrained to a single VC firm’s portfolio. With only 59 nodes and 93 edges, the resulting network is sparse, which necessitates caution when interpreting the stability and generalizability of the detected community structures. While the modularity scores ($Q = 0.472$) suggest a non-random organization, the small sample size means that these partitions may be sensitive to the addition or removal of a few key edges. Consequently, the claims regarding the robustness of the community structure should be viewed as indicative of the 212 portfolio’s specific organization rather than as a universal law of venture capital ecosystems. It would be valuable for future research to expand this analysis to multiple VC firms, which would help determine if these

structural patterns are common across different investment strategies or unique to this specific portfolio.

Another significant limitation of this study is its reliance on public LinkedIn data, which is inherently self-reported and may suffer from survivor bias or outdated records. Such inaccuracies can lead to the omission of critical ties or the inclusion of obsolete affiliations, potentially underestimating the true density of the network or creating spurious communities. For instance, the absence of a specific degree or a past role on a profile may result in a missing edge in the education or experience layers, thereby artificially increasing the modularity of the detected clusters by removing bridging ties. Furthermore, the bipartite nature of the investment layer means that ties between cofounders of different companies are only visible through their shared institutional backgrounds, potentially underestimating the density of the actual, unstructured social network like informal mentorships. To address these gaps, future work could look into combining scraped data with verified corporate registries or direct surveys to get a more complete picture of these professional ties.

Beyond data availability, the study faced significant data formatting and entity resolution challenges. The raw scraped data contained numerous naming irregularities for portfolio companies (e.g., *SOCRadar* vs. *Socradar*) and inconsistent institutional titles, which required extensive string-matching and manual normalization to avoid creating duplicate nodes for the same entity. In addition to that, some profiles were entirely missing or inaccessible due to LinkedIn's privacy settings or account deletions, leading to a slight under-representation of the total possible network size. To improve this, future iterations could employ a more systematic approach to entity resolution, such as using a standardized institutional gazetteer or a more rigorous set of normalization rules to handle common naming variations.

Additionally, specific data integrity challenges were encountered during the scraping of the ground-truth portfolio datasets. These include missing profiles for one investment manager, for whom not any link could be obtained from either the website of 212 or LinkedIn, naming irregularities for portfolio companies across pages, for example, *SOCRadar* and *Socradar*, both of which indicate the same company, and an occasion of a link inconsistency in which the portfolio entry of 212 for *iyzico* provided information for *Insider One* (formerly known as *Insider*) instead. While the current study prioritized a robust automated pipeline for data generation, these anomalies highlight the need for enhanced entity resolution and manual verification in future iterations. A practical next step would be to introduce a human-in-the-loop verification step, such as a simple interface where a researcher can approve or merge suspected duplicate entities (like *iyzico* and *Insider*), so as to catch and fix these kinds of inconsistencies before the network is actually generated.

Another aspect that should be incorporated by prospective future work on this study is temporal data so as to investigate whether networks grow denser or sparser dynamically around funding rounds. In addition, incorporating co-investment data

with entirely separate competitor VCs would illuminate the bridge connectivity across the wider Turkish startup ecosystem at large. Expanding the dataset to include multiple VC firms operating within Turkey would enable a comparative structural analysis of portfolio strategies and institutional biases across the industry.

VIII. CONCLUSION

This study has demonstrated the utility of multilayer network analysis in deciphering the complex social fabric governing venture capital portfolios. By synthesizing institutional and financial data from the 212 ecosystem, a highly structured, disassortative topology ($r = -0.718$) was identified, where investment managers function as critical structural bridges between isolated founder communities. The findings of the study reveal that these communities are not random aggregations but are systematically reinforced by pre-existing institutional pathways, with local alma mater networks—specifically Boğaziçi, METU, and ITU—serving as primary catalysts for trust and connectivity. Conversely, international institutional bridges like LSE and Harvard provide the structural holes necessary for cross-border scalability.

Within the specific context of the 212 portfolio, the results suggest that professional social capital is closely associated with founder visibility and connectivity. Although the dataset is limited to a single venture capital firm and a relatively small sample of actors, the reproducible methodology provides a framework for examining how institutional relationships shape startup ecosystems. More broadly, the findings illustrate how shared educational and professional histories may contribute to trust formation and the emergence of formal investment relationships, while motivating future comparative studies across larger and more diverse venture capital networks.

ACKNOWLEDGMENTS

This study was conducted as a course project for CS 414/514 (Network Science) at Sabancı University. The author would like to thank the instructor for their guidance. The full dataset and code for this project are available at the GitHub repository `OnurOrman/seed-to-scale`.

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APPENDIX A GRAPH VISUALIZATIONS REFERENCE

In addition to the primary figures included throughout the text, a comprehensive suite of computational layouts covering all isolated thematic and bipartite structural components natively generated via ForceAtlas2 and Yifan Hu algorithms has been systematically exported. These reference variants can be found in the `gephi_viz/exported/` directory. Key architectural mappings include:

- **Financial Connectivity:** `vc_company_cofounder`, `vc_company_inv_manager`, `company_cofounder_inv_manager`.
- **Institutional Trust Pathways:** `education`, `experience`, `volunteering`.
- **Aggregated Ecosystem Layouts:** `cofounders_employee` and `all_networks_combined`.

APPENDIX B SUPPLEMENTAL STATISTICAL TABLES

Table III provides a detailed breakdown of the top influencers within the cofounder-investment manager network, highlighting the brokerage role of 212 partners.

TABLE III: Top Brokerage and Influence Rankings (Main Network)

Node Name	Type	Degree	Betweenness
Caglar Urcan	Manager	16	0.270
Selma Bahcivanoglu	Manager	13	0.215
Ali Karabey	Manager	13	0.211
Gizem Yagiz	Manager	10	0.108
Kivanc Aydin	Manager	10	0.137
Batu Sat	Cofounder	7	0.230
Sean Yu	Cofounder	6	0.130
Numan Numan	Manager	6	0.162
Cagdas Yildiz	Manager	6	0.094
Maheer Hakim	Manager	6	0.105

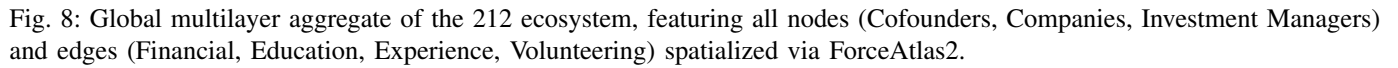
The following clusters (Table IV) were identified using the Louvain algorithm ($Q = 0.472$), showing the institutional focus of each sub-group within the 212 portfolio.

TABLE IV: Characterization of Detected Ecosystem Communities (Full Partition, $N = 59$)

ID	Size	Primary Broker	Defining Signal / Shared Background
1	7	Caglar Urcan	Deep-Tech / Boğaziçi Engineering Hub
2	9	Selma Bahcivanoglu	Travel & SaaS / METU & ITU Pathways
3	8	Ali Karabey	International Scaling / LSE & Harvard
4	11	Gizem Yagiz	Regional Logistics / Local Professional Ties
5	12	Numan Numan	FinTech & Payments / Mixed Professional
6	10	Kivanc Aydin	AI & Data Science / METU Alumni
Residual nodes: 2 nodes not strongly affiliated with any dominant community signal			

APPENDIX C SUPPLEMENTAL COMPUTATIONAL TOPOGRAPHY

Fig. 8 showcases the absolute structural complexity of the 212 ecosystem when all institutional and financial layers are aggregated into a single multilayer projection. Fig. 9 reveals the tripartite core, while Fig. 10 illustrates the scaling properties of average degree and density across these isolated layers.



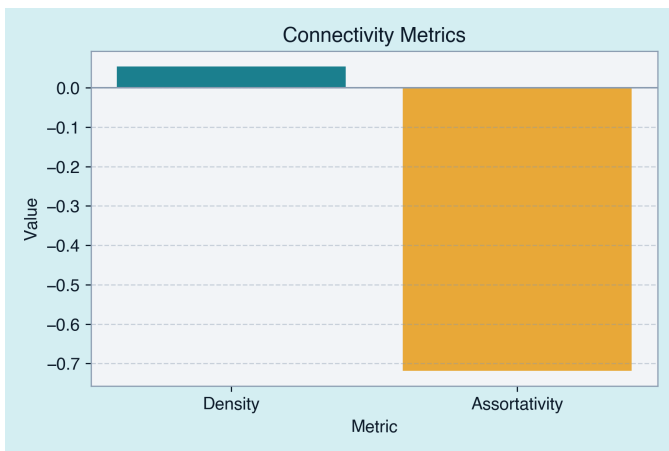


Fig. 10: Comparative analysis of average degree and network density across specific institutional layers, demonstrating the structural density of the professional experience tier relative to education and volunteering.